

# Notes 7

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## Wildcards

### 1. The Asterisk (\*)

The asterisk represents zero or more characters. It is the most flexible wildcard and is typically used when you know how a name starts or ends, but not the middle or the full length.

**Example 1 (File Search):** `ls *.jpg`

Matches every file that ends in `.jpg`, regardless of the filename (e.g., `sunset.jpg`, `photo.jpg`, or `a.jpg`).

**Example 2 (Word Search):** `apple*`

Matches "apple", "apples", "applesauce", or "applecart".

**Example 3 (Database/SQL):** `SELECT * FROM Users WHERE Name LIKE 'Sm%';`

Finds all users whose names start with "Sm", such as "Smith", "Smyth", or "Small".

### 2. The Question Mark (?)

The question mark represents exactly one character. Use this when you know the exact length of the string but are unsure of a specific character.

**Example 1 (File Search):** `ls file?.txt`

Matches `file1.txt` or `fileA.txt`, but will not match `file10.txt` because that has two characters where the question mark is.

**Example 2 (Data Filtering):** `d?g`

Matches "dog" or "dig", but not "drug".

**Example 3 (System Commands):** `rm backup_?.zip`

Matches files like `backup_01.zip` or `backup_ab.zip`, ensuring you only delete files with a two-character suffix.

### 3. Square Brackets ([ ])

Square brackets are used to match any one character inside the brackets. You can specify a list of characters or a range.

**Example 1 (Specific Set):** `ls [ba]at.txt`

Matches `bat.txt` and `aat.txt`, but ignores `cat.txt`.

**Example 2 (Ranges):** `image[1-3].png`

Matches `image1.png`, `image2.png`, and `image3.png`.

**Example 3 (SQL/Regex):** `WHERE ZipCode LIKE '9021[0-5]'`

Finds zip codes from 90210 through 90215.

## Brace Expansion

Brace expansion is a powerful feature in shells like Bash and Zsh that allows you to generate arbitrary strings. When combined with the `mkdir -p` command, it becomes a high-speed tool for building complex, nested directory structures in a single line. The syntax uses curly braces `{ }` to contain a comma-separated list of items or a range.

### The Syntax Basics

- List: `{item1,item2,item3}`
- Range: `{1..5}` or `{a..z}`
- Nesting: `{dir1/{subdir1,subdir2}}` The `-p` (parent) flag in `mkdir` is essential here because it tells the system to create any necessary parent directories that don't already exist.

### Example 1: Project Scaffolding

If you are starting a new software project, you often need a standard set of folders for source code, documentation, and testing.

- The Command: `mkdir -p project/{src,docs,tests,bin}`

### Example 2: Nested Year and Month Structure

If you are organizing data or logs by date, you can nest expansion sets inside one another to create a deep tree instantly.

- The Command: `mkdir -p archive/202{4..6}/{01..12}`